Calibrating Maximum Cross-Correlation Tests Using Independent Moving Block Resampling

Zhixun (Ricardo) Zheng
Independent Researcher

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Abstract

Selecting the largest absolute sample cross-correlation over candidate lags can inflate the Type I error rate, particularly when each series is serially dependent. We apply independent marginal moving block resampling to calibrate the maximum statistic under the null that the two processes are independent. Every bootstrap replication repeats the full lag scan, accounting for the search while approximately retaining short-range dependence within each series.

In a Gaussian AR(1) simulation with N = 300, maximum lag L = 10, and 1000 Monte Carlo trials, an unadjusted IID max-correlation sweep rejects in 81.2% of null trials at a nominal 5% level. Moving block calibration reduces this proportion to 6.4%, substantially reducing the distortion while leaving a rejection rate above the nominal level. Under a specified lagged alternative with coupling coefficient $\beta = 0.25$, the rejection proportion is 76.5%.

1 Introduction

Cross-correlation analysis is commonly used to explore temporal associations between stationary time series [1]. Researchers often scan a range of candidate lags and retain the largest absolute correlation. A single-lag critical value does not account for this search: the reported statistic is a maximum over correlated quantities, with a different null distribution from a correlation at one prespecified lag. Serial dependence within either series can further invalidate an IID threshold.

Olden and Neff [2] examine the bias caused by searching across lags and distinguish it from the effects of temporal autocorrelation. Block bootstrap methods provide a way to retain local serial dependence when approximating sampling distributions [3, 4]. Independently resampling the marginal series has also been used to construct reference distributions for time-series independence tests [5].

We evaluate independent marginal block resampling for calibrating the maximum absolute sample cross-correlation. Every bootstrap replication repeats the complete lag scan. In a controlled Gaussian AR(1) simulation, we compare its rejection rates with those of three IID-based diagnostic baselines.

2 Problem Setup

Let $\{(X_t, Y_t)\}_{t\in\mathbb{Z}}$ be a jointly stationary real-valued process with finite, positive marginal variances, observed at $t = 1,\dots,N$. For a prespecified maximum lag L, define the candidate set $\mathscr{L} = \{-L, -L+1, \dots, L\}$ and the population cross-correlation

$$\rho(l) = \mathrm{Corr}(X_t, Y_{t+l}), \qquad l \in \mathscr{L}. \tag{1}$$

At lag l, the available paired observations are indexed by

$$I_l = \{t : 1 \le t \le N,\ 1 \le t+l \le N\}.$$

The sample correlation $\hat{\rho}(l)$ is the ordinary Pearson correlation of $\{(X_t, Y_{t+l}) : t \in I_l\}$, using means and variances calculated over that lag-specific overlapping sample of size $N - |l|$. The test statistic is

$$T_{\mathrm{obs}} = \max_{l\in\mathscr{L}} |\hat{\rho}(l)|. \tag{2}$$

Positive l pairs X_t with a later value of Y.

We calibrate this statistic under the strong null of independence between the two processes:

$$H_0 : \{X_t\}_{t\in\mathbb{Z}} \perp \{Y_t\}_{t\in\mathbb{Z}}. \tag{3}$$

This null implies $\rho(l) = 0$ at every candidate lag. The statistic targets linear cross-series association within the searched lag range; it need not detect nonlinear dependence with zero cross-correlations.

The maximum in (2) is at least as large as the absolute correlation at any one candidate lag. Its null distribution therefore differs from the distribution used by a single-lag test. Serial dependence additionally changes the variability and covariance of the sample correlations, so correcting only the number of comparisons need not yield valid calibration.

3 Methodology

3.1 Independent Moving Block Resampling

For a block length K, form the overlapping marginal blocks

$$B_i^X = (X_i, \dots, X_{i+K-1}), \qquad B_i^Y = (Y_i, \dots, Y_{i+K-1}), \tag{4}$$

for $i = 1,\dots,N-K+1$. For each bootstrap replication b, sample $\lceil N/K \rceil$ blocks with replacement from the X collection, and independently sample the same number from the Y collection. Concatenate each block sequence and truncate it to length N, obtaining $X^{*(b)}$ and $Y^{*(b)}$.

Independent sampling imposes cross-series independence in the conditional bootstrap distribution. Observation order within the blocks approximately retains short-range marginal dependence, although connections across block boundaries are broken. This is the overlapping moving block bootstrap, applied separately to each marginal series [3, 4].

3.2 Calibration of the Lag Scan

Apply the same lag set and overlapping-sample correlation calculation to every resampled pair:

$$T^{*(b)} = \max_{l\in\mathscr{L}} |\hat{\rho}^{*(b)}(l)|, \qquad b = 1,\dots,B. \tag{5}$$

The bootstrap distribution approximates the null distribution of the maximum, including the lag search. We estimate its upper-tail probability using the plus-one rule:

$$\hat{p} = \frac{1 + \sum_{b=1}^{B} \mathbf{1}\{T^{*(b)} \geq T_{\text{obs}}\}}{B+1}. \tag{6}$$

The decision rule is $\hat{p} < \alpha$. With $B = 999$ and $\alpha = 0.05$, a value of exactly 0.050 is not rejected. The plus-one rule avoids a zero estimated tail probability; calibration still depends on how well the block bootstrap approximates the relevant null distribution.

4 Simulation Study

4.1 Data-Generating Processes and Configuration

Under H_0, generate mutually independent Gaussian AR(1) series:

$$X_t = \phi_X X_{t-1} + \eta_{t,X}, \tag{7}$$

$$Y_t = \phi_Y Y_{t-1} + \eta_{t,Y}, \tag{8}$$

where both innovation sequences are independent standard Gaussian white noise. Under the alternative, retain the same X process and construct

$$Y_t = \beta X_{t-l_0} + u_t, \tag{9}$$

$$u_t = \phi_Y u_{t-1} + \eta_{t,Y}, \tag{10}$$

with the innovation sequence for u independent of that for X. This adds a lagged linear signal to autocorrelated noise.

The main experiment uses $N = 300$, $L = 10$, $\phi_X = 0.6$, $\phi_Y = 0.5$, $l_0 = 2$, and $\beta = 0.25$. Recursions start at zero, and a burn-in of 500 observations is discarded. There are $R = 1000$ Monte Carlo trials under each regime, with $B = 999$ bootstrap replications per MBB test. Both marginal resamplers use

$$K = \lceil N^{1/3} \rceil = 7.$$

This is a rate-based block-length heuristic, not an optimized choice for the statistic. Simulations use Python and NumPy [6]. For zero-based trial index i, the random seed is $2026 + i$ under H_0 and $2026 + i + 9999$ under H_1.

4.2 Metrics and Diagnostic Baselines

Empirical size is the rejection proportion under H_0 at $\alpha = 0.05$. We also report the rejection proportion under the specified alternative. Four procedures are evaluated:

- **Naive max-correlation sweep.** Reject if $T_{\text{obs}} > 1.96/\sqrt{N}$. This IID single-lag threshold ignores the search and serial dependence.
- **Fixed-lag IID oracle.** Evaluate only $|\hat{\rho}(l_0)|$ and use the same threshold. The lag is prespecified and matches the signal under H_1; serial dependence remains unaccounted for.
- **Holm (IID), global decision.** For each lag, calculate the two-sided normal-approximation IID value

  $$p_l = 2\{1 - \Phi(\sqrt{N}\,|\hat{\rho}(l)|)\}.$$

  The global decision records whether Holm's procedure [7] rejects at least one lag. This occurs if $\min_l p_l < \alpha/(2L+1)$, and coincides with the Bonferroni global decision. The implementation uses $N = 300$ at every lag, even though the correlation samples have size $N - |l|$.

- **MBB.** Calibrate the observed maximum using independent marginal block resampling and (6).

The first three procedures are IID diagnostic baselines, so their rejection rates should not be treated as an equal-size comparison with MBB.

4.3 Results

Figure 1 shows the first null trial. Its largest absolute correlation is $T_{\text{obs}} = 0.1335$, exceeding the IID threshold $1.96/\sqrt{300} \approx 0.1132$. Thus, the naive sweep rejects despite the independence of the generating processes.

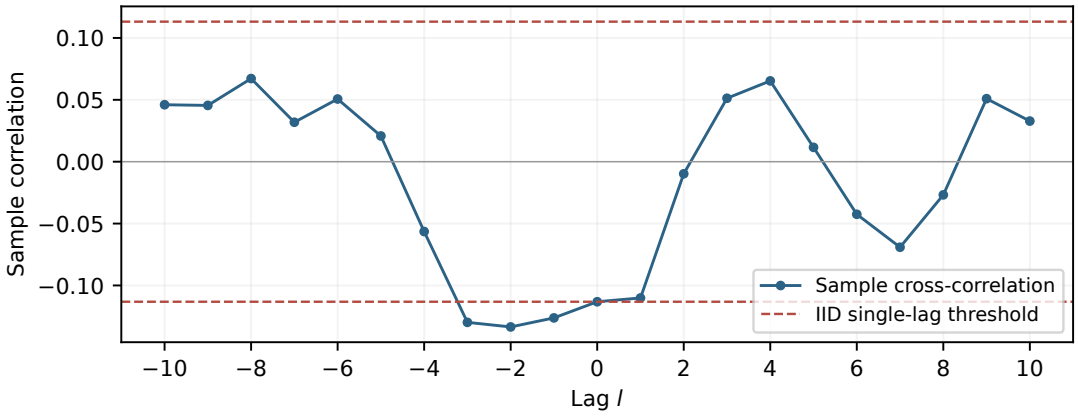


Figure 1: Lag scan for the first independent AR(1) null trial. The dashed lines mark the IID single-lag threshold.

Table 1 and Figure 2 report rejection proportions across the 1000 trials per regime. Under H_0, the naive sweep rejects in 81.2% of trials. Removing the search in the fixed-lag IID oracle reduces the rate to 14.2%, which is still above 5%. Holm's global decision yields 23.7%: adjusting multiplicity does not repair the misspecified IID marginal values.

Table 1: Rejection proportions in the reported configuration, with 1000 Monte Carlo trials per regime.

Procedure	Under H_0	Under H_1
Naive max-correlation sweep	0.812	0.997
Fixed-lag IID oracle	0.142	0.972
Holm (IID), global decision	0.237	0.907
MBB	0.064	0.765

The MBB procedure rejects in 64 of 1000 null trials, giving an empirical size of 0.064 and an exact 95% binomial interval of $[0.0496, 0.0810]$. It substantially reduces the size distortion, although the observed rejection rate remains above the nominal 0.05 level. Under H_1, it rejects in 765 of 1000 trials, with an exact 95% binomial interval of $[0.7375, 0.7910]$. These intervals describe Monte Carlo uncertainty in the rejection probabilities.

The IID-based procedures have larger rejection proportions under H_1, but their null rejection rates are also substantially inflated. These results therefore do not establish a power ranking at a common controlled size. For the illustrative trial, the MBB tail estimate is $\hat{p} = 0.608$, so the null is not rejected. Figure 3 shows its bootstrap distribution with a 95th percentile of 0.2152; Appendix A reports the running percentile estimate.

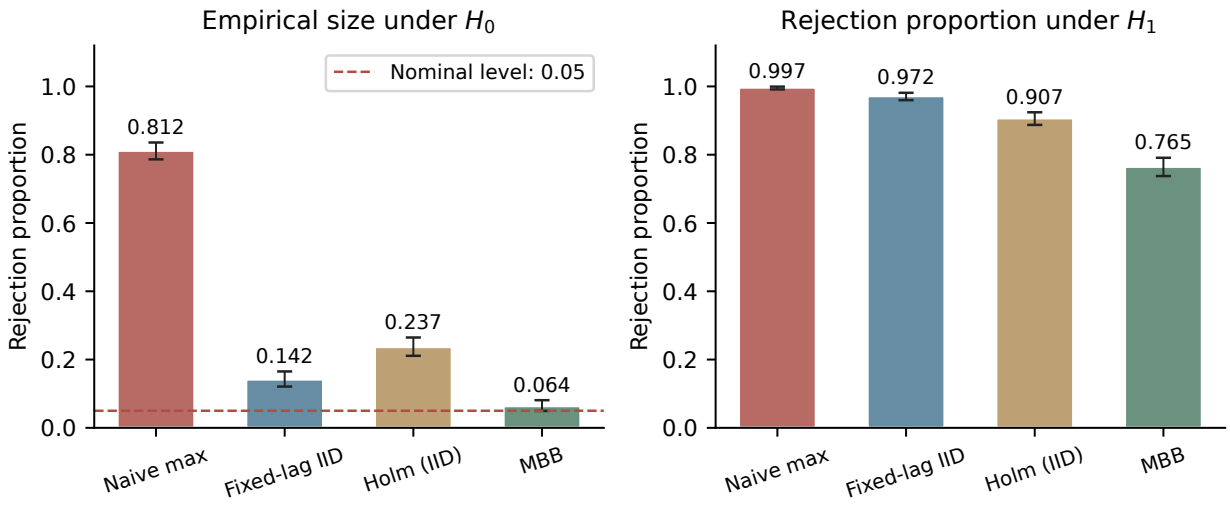


Figure 2: Rejection proportions under the null and alternative. Error bars are exact 95% binomial intervals; the dashed line marks the nominal 0.05 level.

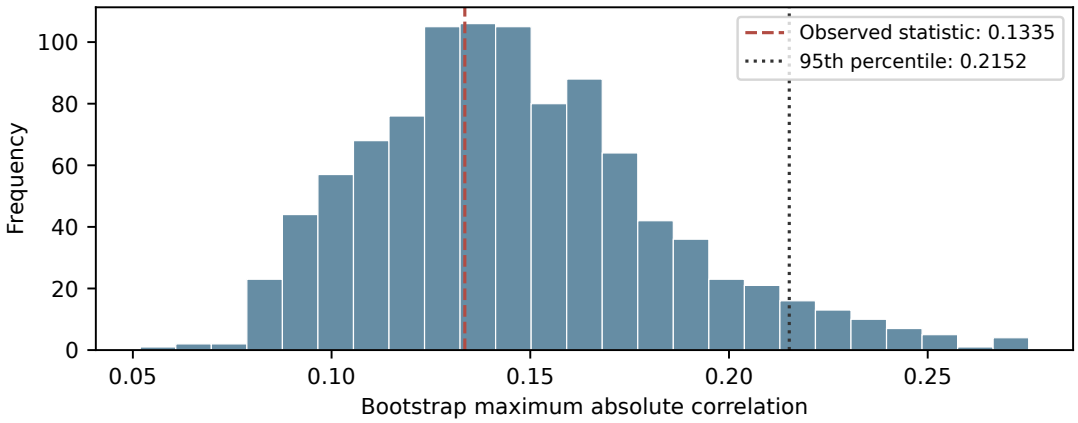


Figure 3: Bootstrap maximum-statistic distribution for the first null trial. The observed statistic is 0.1335; the empirical 95th percentile is 0.2152.

5 Scope and Limitations

The experiment considers one Gaussian AR(1) configuration under process independence and one specified lagged alternative. We use K = 7 throughout; the effects of block length, stronger serial dependence, sample size, and lag-scan width remain untested. Automatic block-length methods [8, 9] provide a possible extension.

The independence null excludes contemporaneous and lagged cross-series dependence. Independent marginal resampling therefore cannot be used for a null allowing contemporaneous coupling.

Appendix B illustrates how contemporaneous coupling and persistence can produce correlations at nonzero lags. A rejection therefore does not identify a direct lagged effect or causal direction. A null that allows contemporaneous coupling would require a resampling scheme matched to that structure.

6 Conclusion

In the reported AR(1) experiment, independent marginal block resampling reduced the null rejection rate from 81.2% for the naive IID lag sweep to 6.4%, against a nominal 5% level. Under the specified lagged alternative, the calibrated test rejected 76.5% of trials. These results show why calibrating a maximum cross-correlation requires attention to both lag selection and serial dependence.

Declaration of Generative AI Use

Generative AI tools assisted with language editing, literature searches, and manuscript structuring.

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A Bootstrap Percentile Diagnostic

For the first null trial, Figure 4 plots the empirical 95th percentile calculated from the first b bootstrap statistics, for $b = 10, \ldots, 999$. Percentiles use NumPy's default linear interpolation. The estimate is more variable with few draws and ends at 0.2152. This describes Monte Carlo variability for one fixed dataset; the test decisions use (6).

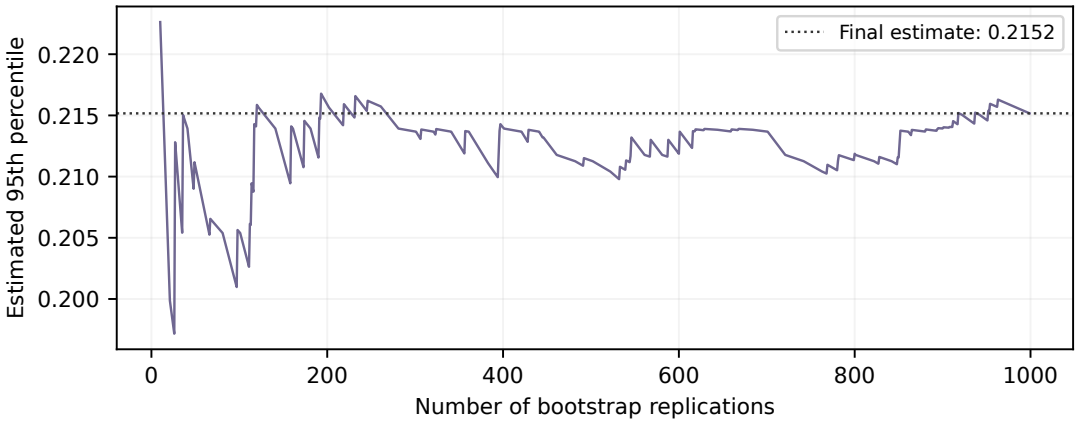


Figure 4: Running estimate of the bootstrap 95th percentile for the first null trial.

B Contemporaneous Coupling and Nonzero-Lag Correlation

Consider mutually independent Gaussian AR(1) processes

$$X_t = 0.6X_{t-1} + \varepsilon_{t,X}, \tag{11}$$

$$U_t = 0.5U_{t-1} + \varepsilon_{t,U}, \tag{12}$$

with independent standard Gaussian innovations, and construct

$$Y_t = -0.3X_t + U_t. \tag{13}$$

The coefficient -0.3 specifies contemporaneous coupling, not the resulting correlation. In the stationary distribution,

$$\mathrm{Cov}(X_t, Y_{t+l}) = -0.3\ \mathrm{Var}(X_t)\, 0.6^{|l|}. \tag{14}$$

Thus, persistence in X generates cross-correlation at nonzero lags even though the coupling term uses only X_t.

Figure 5 shows one realization of this contemporaneously coupled model. The maximum absolute cross-correlation exceeds the MBB critical value (approximately 0.222), so the independence null is rejected. The dashed lines mark the positive and negative thresholds for the maximum statistic across the lag scan. Since the coupling acts at lag zero, this rejection is not evidence of a direct delayed effect.

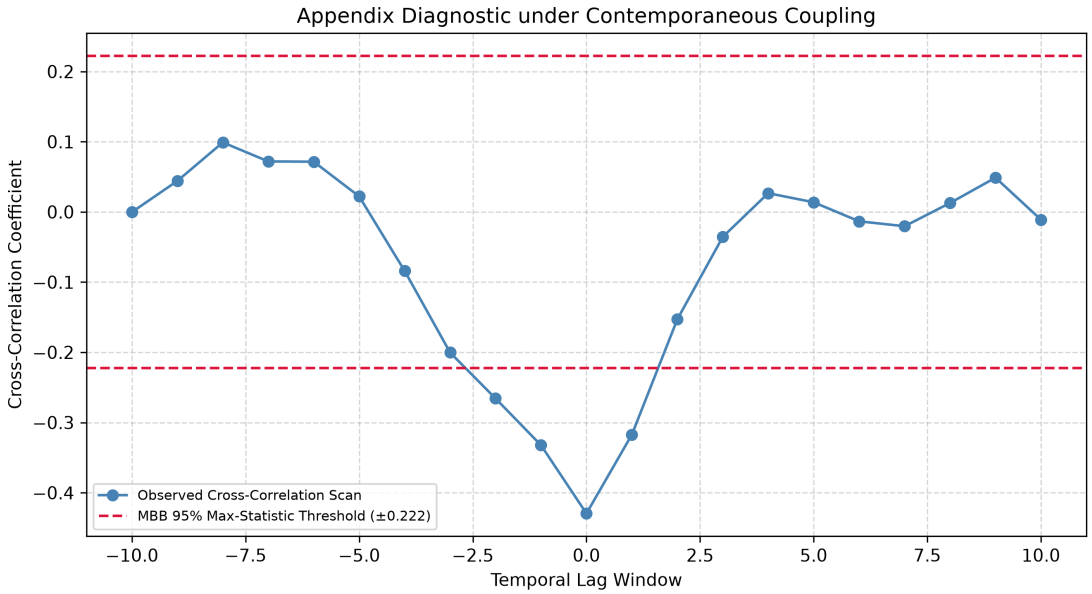


Figure 5: Illustrative scan under $Y_t = -0.3X_t + U_t$. The contemporaneous coupling induces correlation at neighboring lags. Dashed lines show the MBB maximum-statistic threshold (approximately ± 0.222) under process independence.